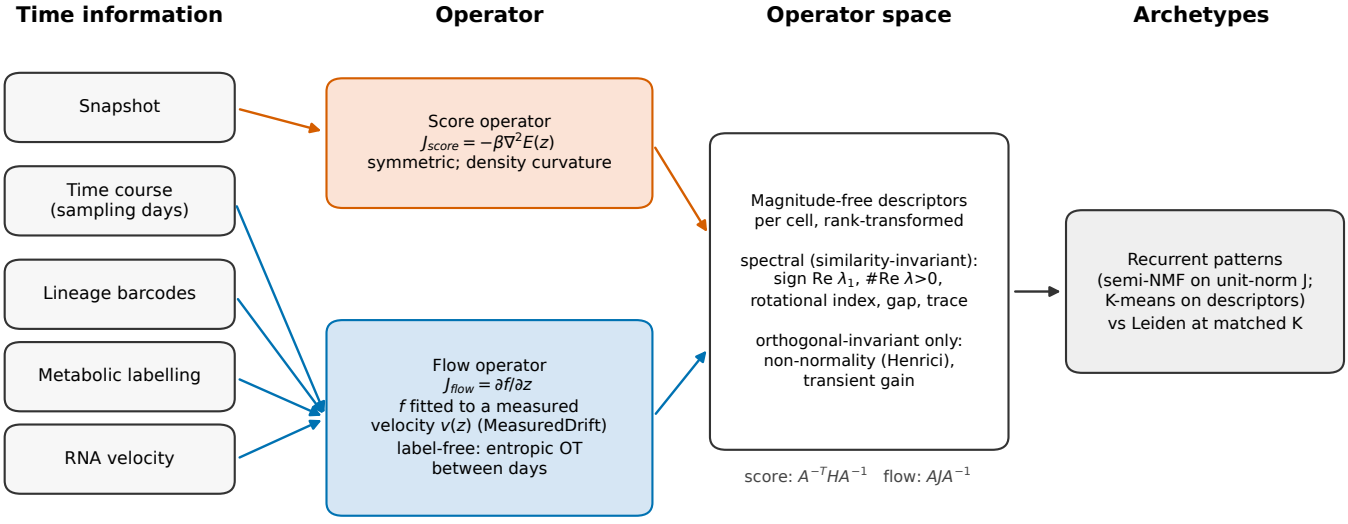
Time information
Operator
Operator space
Archetypes
Snapshot
Time course
(sampling days)
Lineage barcodes
Metabolic labelling
RNA velocity
Score operator
$J_{score} = -\beta\nabla^2 E(z)$
symmetric; density curvature
Flow operator
$J_{flow} = \partial f/\partial z$
f fitted to a measured
velocity $v(z)$ (MeasuredDrift)
label-free: entropic OT
between days
Magnitude-free descriptors
per cell, rank-transformed
spectral (similarity-invariant):
sign Re λ_1, #Re λ>0,
rotational index, gap, trace
orthogonal-invariant only:
non-normality (Henrici),
transient gain
score: $A^{-T}HA^{-1}$ flow: AJA^{-1}
Recurrent patterns
(semi-NMF on unit-norm J;
K-means on descriptors)
vs Leiden at matched K
Identifiability components (each tested in the Results)
Representation
embedding metric
(S12, S13)
Clock
time units, k
(S14)
Rotation
antisymmetric flow
(S21b, S15)
Stationarity
score = stability only
at steady state (S10)
Diffusion
not identifiable at
48-h sampling (S10)
Composition
proliferation/death,
occupancy (S18b)